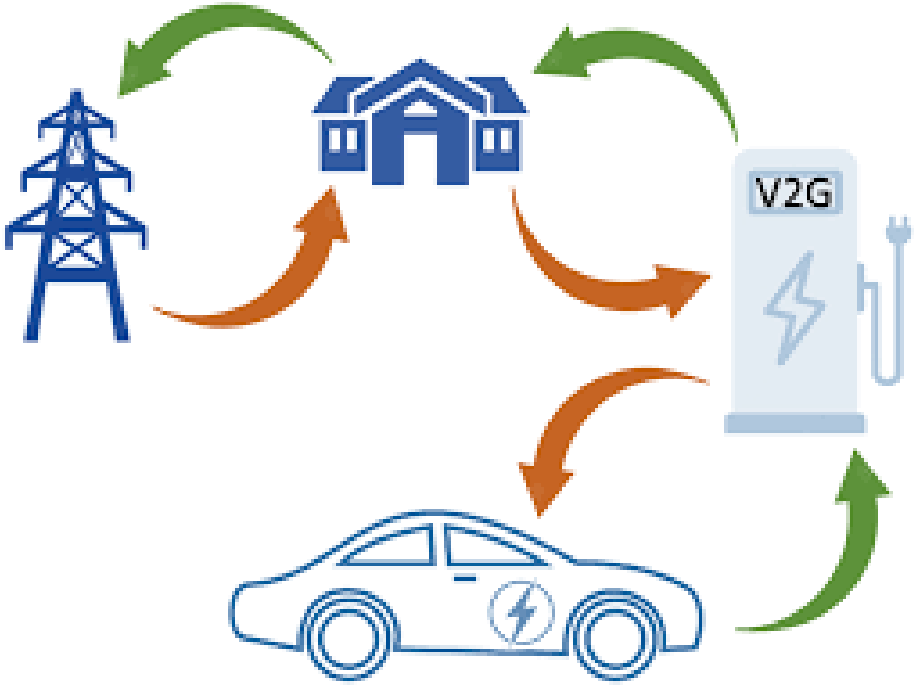


Vehicle to Grid & Grid to Vehicle

EEE 306
Sec: C1
Group: 02
Id:
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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 306

Power System I Laboratory

Final Project Report

Section: C1 Group: 02

Vehicle to Grid & Grid to Vehicle

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Signature of Instructor: _____

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

Signature: _____ **Full Name: Md Symria Raihan** **Student ID: 2006133**	**Signature:** _____ **Full Name: S.M Razibur Rahman** **Student ID: 2006134**
Signature: _____ **Full Name: Bashir Shahriyer Rafi** **Student ID: 2006136**	**Signature:** _____ **Full Name: Md Amir Hamza** **Student ID: 2006145**
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1 Abstract

This project explores the transformative potential of Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) technologies in the realm of electric vehicles (EVs) and energy grids. V2G technology leverages the capabilities of EV batteries to discharge stored energy back to the grid during peak demand periods, thereby aiding grid stability and reducing electricity costs. Conversely, G2V facilitates the charging of EVs from the grid, allowing for the utilization of renewable energy sources or low-cost electricity. By enabling bi-directional energy flow between EVs and the grid, both V2G and G2V hold promise for enhancing grid resilience, supporting renewable energy integration, and optimizing EV charging. This report delves into the underlying principles, benefits, challenges, and potential applications of V2G and G2V technologies, presenting a comprehensive overview of their significance in the transition towards a sustainable energy future.

2 Introduction

In the quest for sustainable energy solutions, the intersection of transportation and power systems has emerged as a promising frontier. Electric vehicles (EVs), once viewed solely as a means of reducing carbon emissions in transportation, are now recognized as potential game-changers in the broader energy landscape. One of the most intriguing developments in this domain is the concept of Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) technologies, which enable bidirectional energy flow between EVs and the electrical grid. These technologies hold immense potential to revolutionize energy management practices, particularly within micro-grid contexts.

This project sets out to explore and advance the understanding of V2G and G2V technologies within the framework of micro-grid systems. Microgrids, characterized by their localized energy generation, distribution, and consumption networks, are increasingly gaining traction as resilient and sustainable alternatives to traditional centralized power grids. However, their efficacy is often contingent upon efficient energy storage and management solutions.

EVs, with their large-scale deployment and proliferation, present a unique opportunity to address some of the key challenges faced by microgrids. By harnessing the energy stored in EV batteries, microgrids can enhance their resilience, optimize energy utilization, and facilitate the integration of renewable energy sources. The concept is simple yet transformative: EVs serve as mobile energy storage units, capable of storing surplus energy during periods of low demand (G2V) and feeding it back into the grid during peak demand (V2G).

The crux of this project lies in designing and implementing a V2G-G2V system within a micro-grid environment, leveraging DC fast charging architecture. DC fast charging, known for its rapid charging capabilities, is particularly well-suited for micro-grid applications where quick power transfer is essential for maintaining grid stability. By interfacing EVs with the micro-grid through off-board chargers and grid-connected inverters, this project seeks to demonstrate the feasibility and effectiveness of integrating EVs into micro-grid energy management strategies.

The journey begins with a thorough review of existing literature and research in the field, examining previous studies on V2G and G2V technologies, as well as their applications within micro-grid contexts. Drawing upon this knowledge, the project proceeds to develop a comprehensive architectural framework for implementing V2G-G2V systems, incorporating the latest advancements in DC fast charging technology and control systems.

Simulation studies form a critical component of the project, providing insights into the performance, efficiency, and scalability of the proposed V2G-G2V system. Through detailed modeling and analysis, the project aims to validate the efficacy of the system under various operating conditions, shedding light on its potential benefits and limitations. The simulations were done in MATLAB Simulink. The grid was modelled by a synchronous generator. There were several loads connected to the generator.

To model the variable demand of the load we used switching techniques. And to incorporate this with the charging and discharging of the vehicle modelled by the battery we used modulation techniques. The simulation outputs were able to accurately capture the varying demands of the grid and charging and discharging of the battery.
Ultimately, this project aspires to contribute to the growing body of knowledge surrounding V2G and G2V technologies, offering valuable insights into their practical implementation within micro-grid environments.

3 Design

3.1 Problem Formulation

3.1.1 Literature review

- **Historical Overview:** The concept of vehicle-to-grid (V2G) and grid-to-vehicle (G2V) power supply has evolved alongside the development of electric vehicles (EVs) and smart grid technologies. The idea of using EV batteries to store and supply electricity to the grid emerged in the early 2000s as a way to improve grid stability, reduce peak demand, and integrate renewable energy sources. Early research and pilot projects laid the foundation for standards and regulations governing V2G and G2V systems, paving the way for their commercial deployment.

- **Working Principles:** Vehicle-to-grid (V2G) systems enable bi-directional power flow between EV batteries and the electrical grid. When plugged into a V2G charging station, an EV can either charge its battery from the grid or discharge electricity back to the grid, depending on demand and pricing signals. Grid-to-vehicle (G2V) systems, on the other hand, allow EVs to charge their batteries from the grid, typically during off-peak hours when electricity is cheaper and grid demand is lower.

- **Design Parameters:** V2G and G2V systems require compatible charging infrastructure and communication protocols to facilitate bi-directional power flow. V2G chargers must be capable of converting grid AC power to DC for battery charging and converting DC power from the battery to AC for grid supply. G2V chargers perform the reverse operation, converting AC power from the grid to DC for battery charging.

- **Applications:** Vehicle-to-grid (V2G) and grid-to-vehicle (G2V) power supply systems have the potential to revolutionize the energy landscape by turning EVs into mobile energy storage units. These systems can help balance supply and demand on the grid, reduce the need for expensive grid upgrades, and increase the use of renewable energy sources. V2G and G2V technologies are being piloted and deployed in various regions around the world, with ongoing research focused on optimizing their performance and integration with existing grid infrastructure.

-

3.2 Design Method

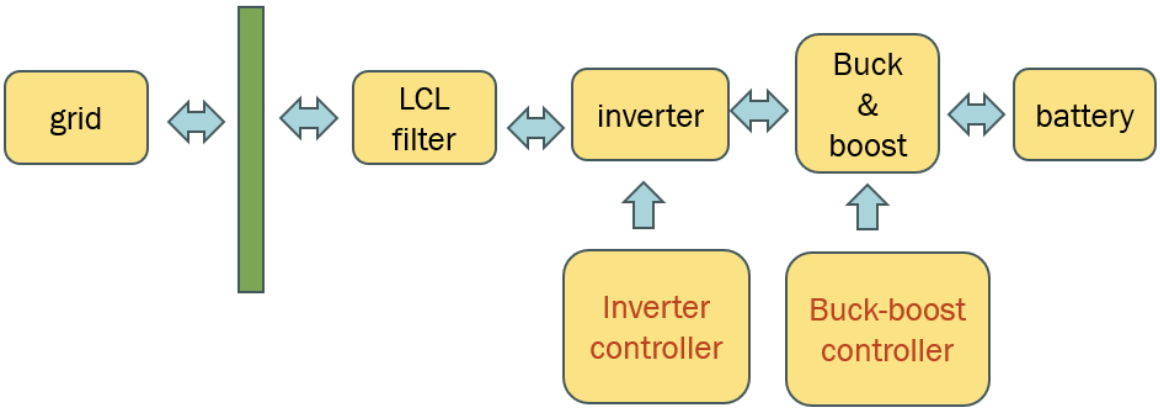


Figure 1 Design Method

3.3 Circuit Diagram

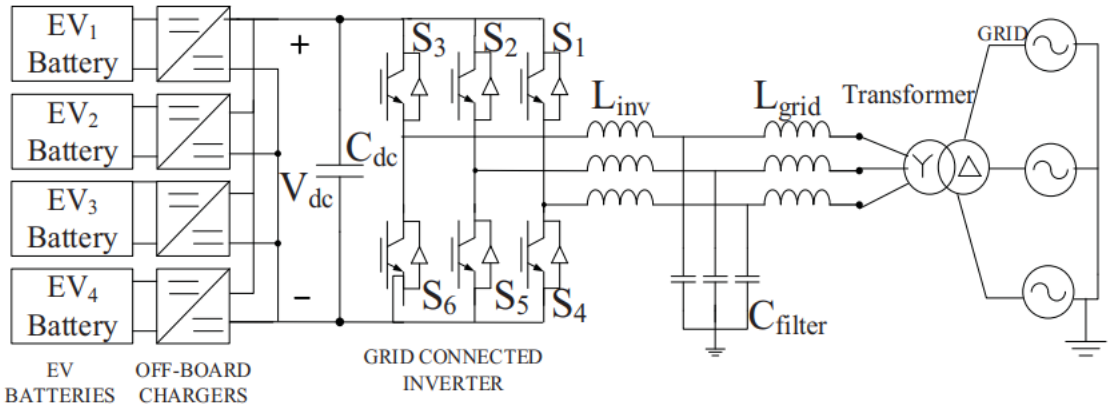


Figure 2 Circuit Diagram

The following sub-systems are included in our project:

- **EV Battery:** In Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) projects, Electric Vehicle (EV) batteries serve as pivotal components, facilitating bidirectional power flow between the vehicle and the grid. These batteries store electrical energy, acting as buffers that can absorb excess energy from renewable sources or off-peak grid electricity for later use or injection back into the grid. They provide power to drive the vehicle's electric motor and can also supply power to the grid during peak demand or other grid needs, supporting bidirectional power flow. The Battery Management System (BMS) plays a crucial role in monitoring and controlling the battery's state of charge, temperature, and overall health to ensure safe and efficient operation. By participating in V2G programs, EV batteries can provide grid services such as peak shaving, frequency regulation, and voltage support, enhancing grid stability and reliability while offering economic benefits to EV owners or operators.

- **Buck-Boost Converter:** This type of converter is used to regulate the voltage levels between the EV's battery and the grid. It can step up (boost) or step down (buck) the voltage as needed to ensure efficient power transfer. It actually determines that the EV Battery will consume or provide power.

- **Inverter:** This component converts the direct current (DC) from the EV's battery into alternating current (AC) that can be fed back into the grid. In V2G mode, the inverter allows the EV to act as a power source for the grid. At the time of charging this convertor acts as an AC to DC converter as well.

- **LCL Filter**: In the context of power electronics, an LCL filter is often used to reduce harmonic distortion and improve the quality of the AC output from converters. It helps in maintaining the grid's stability and ensuring compliance with grid codes and standards.

In this project we have used LCL filter instead of LC filter because in this type of work using LC filter's capacitor acts as a load.

The Matlab code that has been used to calculate the desired value of the capacitor and inductor is below:

```
clc;
k=0.9; %Modulation Index
f1=50; %fundamental frequency
fs=10000 %Switching frequency
vo=400/sqrt(3); %per phase
voh_avg=2; %%0.05*400/sqrt(3); Harmonic voltage
io=30; %Nominal Current
Vdc=800; %dc voltage
a=k^2;
b=(15/4)*k^4;
c=(64/(5*pi))*k^5;
d=(5/4)*k^6;
ke=sqrt((a-b+c-d)/1440)
aa=ke*(Vdc/voh_avg);
bb=vo/(io*fs);
cc=f1/fs;
Lf=bb*sqrt(aa*(1+4*pi^2*cc^2*aa))
Cf=ke*Vdc/(Lf*fs^2*voh_avg)
```

The output:

```
ke =

    0.0080

Lf =

    0.0046

Cf =

    3.4017e-05
```

4 Implementation

4.1 Description

Our project aims to design a simple scheme where an electric vehicle will be able to receive power from the grid(G2V) charge its battery during peak hours and return power back to the grid(V2G) when load demand is very high. The entire model is simulated using Simulink. The model can be subdivided into several blocks for a better understanding of the overall workflow. The blocks are:-

- Microgrid (A single generator bus where loads are connected)
- Lithium-ion battery
- Buck-Boost Converter
- 3-phase inverter (Bidirectional)
- LCL filter
- PLL(Phase Lock Loop controller)
- Clark and Park Transformation Block
- Buck-Boost PWM generator
- DC bus voltage controller
- Demodulator

A block diagram of the entire system is shown to get an idea on how the various blocks interact with each other .

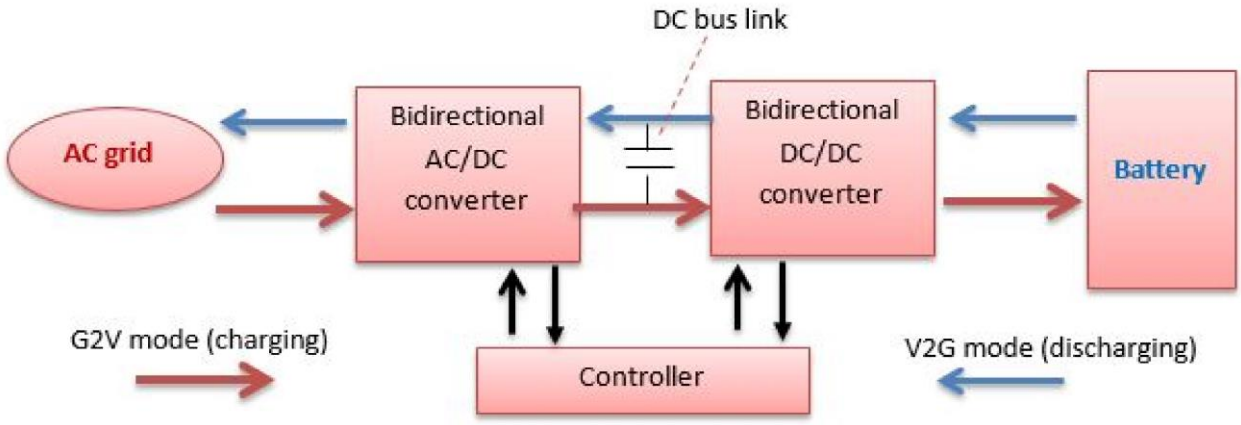
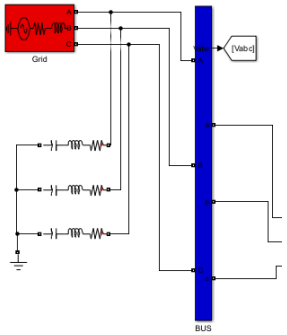


Figure 3 V2G & G2V

Now, we will describe each block in detail.

Microgrid:

We have used a simple generator bus as our grid. Loads are connected to this grid in parallel to each other. A single generator is the source of power in our system.



Rechargeable Battery :

A lithium-ion battery that models an electric vehicle. The voltage specification of this battery is set to 360V, and it step up to 800V using boost converter in order to synchronize it with the grid. During charging period, 30 amperes of current enters the battery and during discharge 30 amperes of current is injected into the grid. The initial state of charge is set to 50% and it decreases during discharge and increases during charging .

Boost Converter:

Used to step up and down the DC voltage of battery. Implemented using IGBT MOSFETs ,capacitors and inductors. The IGBT or MOSFET acts as a switch, cycling between on and off states. During the on state, current flows through the inductor, storing energy. In the off state, the inductor discharges, transferring energy to the output. The inductor smoothens the current flow and stores energy when the switch is on. When the switch turns off, the inductor releases this stored energy, causing an increase in output voltage. The value at which it steps the voltage up to is decided via the duty cycle of the PWM signal that controls the switching of the MOSFETS. The relation is given by,

$$Vdc = \frac{Vbat}{1-D'}$$

Buck Converter:

Performs the opposite of boost converter meaning it steps down the dc voltage for battery. The operation of the buck converter is based on the principle of storing energy in an inductor. The voltage drop across an inductor is proportional to the change in the electric current flowing through the device. A switching transistor is used in between input and output for continuous switches on and off at high frequency. To maintain a continuous output, the circuit uses the energy stored in the inductor. The controlled switch is turned on and off by using PWM (Pulse Width Modulation). PWM can be time-based or frequency based. Time-based Modulation is mostly used for buck converters because it is simple to construct and use. The frequency remains constant in this type of PWM modulation. The relation governing the voltage output of this converter and battery voltage is given by,

$$Vbat = Vbat * D \;\; ; Where\ D = duty\ cycle\ of\ PWM\ signal$$

If we stack both converters one upon another, then we get a buck-boost converter which is a bi-directional converter. The schematic of the circuit is given below:-

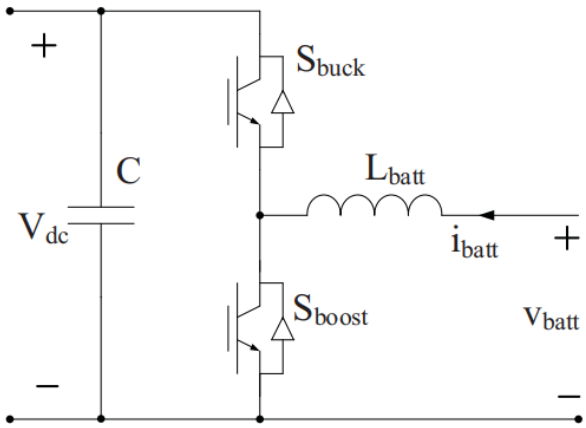


Figure 4 Buck Boost Converter Circuit

When the upper switch is operating, the converter acts as a buck converter stepping down the input voltage to battery charging voltage. During the on state, current flows through the switch and inductor to the battery. This is the charging operation, where the power flow is from the grid to vehicle (G2V). When the switch is off, the current takes its return path through the inductor and diode of lower switch and completes the circuit. When the lower switch (Sboost) is operating, the converter acts as a boost converter stepping up the battery voltage(Vbat) to the dc bus voltage(Vdc). When the switch is in on state, current continues to flow through the inductor and completes its circuit through the anti-parallel diode of the upper switch, and the capacitor. The net power flow in this case is from the vehicle to the grid (V2G) and the battery operates in the discharge mode.

3-Phase Inverter

Inverter Mode

The basic operation of an inverter is to convert AC into DC. It consists of 6 IGBT MOFETS that act as high frequency switches. The PWM signals are generated in such a fashion that these switches are turned on and off in pairs such that the phase voltage is alternating in nature though it may not be perfect sinusoid but contains high frequency harmonics that will be filtered out using LCL filter. The circuit schematic is shown below:-

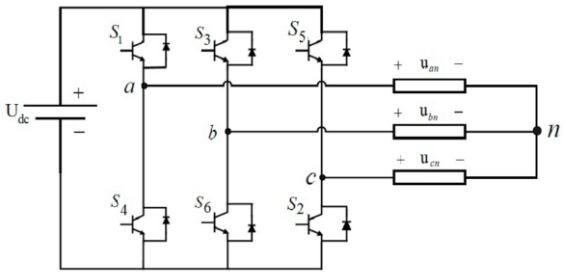


Figure 5 Inverter

To understand its operation, we can replace each MOSFET with a simple switch. The simplified circuit is shown below:-

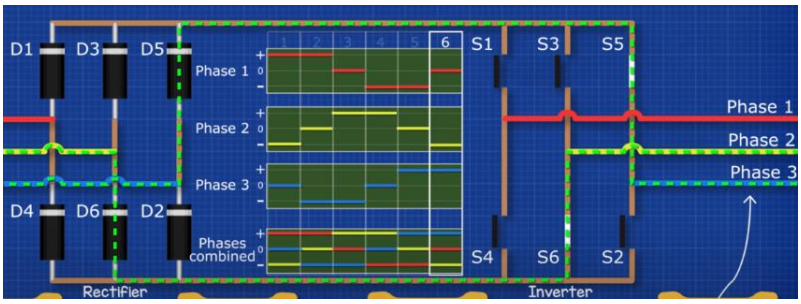


Figure 6 Equivalent Circuit of Inverter

One possible switching sequence is listed below:-

	Switch	State	Switch	State
1	S1,S6	On	All others	Off
2	S1,S2	On	All others	Off
3	S3,S4	On	All others	Off
4	S4,S5	On	All others	Off
5	S3,S2	On	All others	Off
6	S5,S6	On	All others	Off

Active Rectifier Mode (Front End Rectifier)

The 3-phase inverter can also be used as a rectifier during V2G operation. The switches can be turned on and off in such a fashion that current can flow in a certain direction only. Two different sets of PWM signals are generated from the controller system based on whether current crosses a certain threshold or reference or not.

Control Systems:

1. ***PLL (Phase Loop Lock) Control Block:***

A Phase-Locked Loop (PLL) is an electronic feedback system that helps synchronize the phase and frequency of an output signal with a reference signal. PLLs are widely used in various applications, including communication systems, clock generation, frequency synthesis, and audio systems. The block diagram of a PLL controller is shown below: -

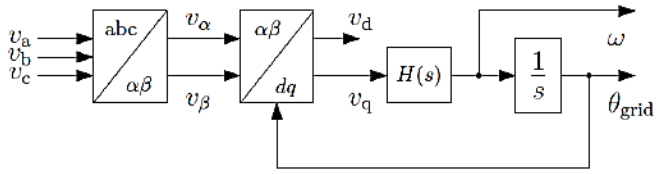
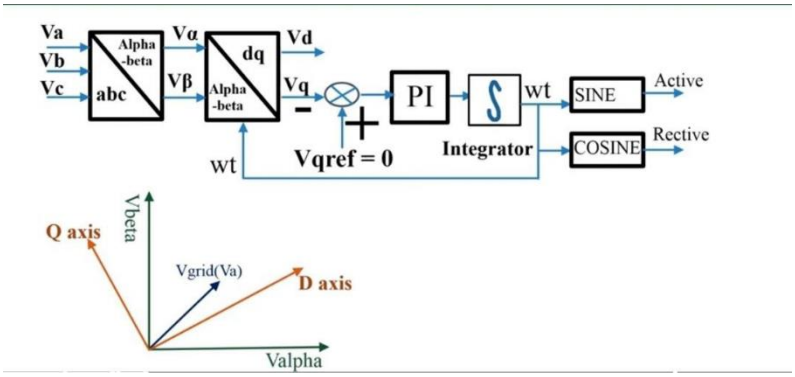
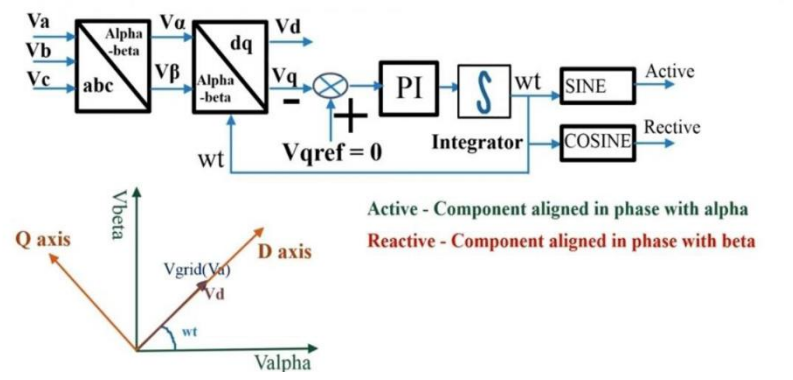


Figure 7 PLL Controller

The PLL takes in a reference input signal, which is the desired signal that we want to synchronize with. In our case, the reference input is the active current component (IQ component) which is set to zero. This is necessary to synchronize or lock in the dq frame with the grid voltage. This is shown in the following figure.





In summary, a PLL is a control system that adjusts the frequency and phase of an output signal to match a reference signal, ensuring synchronization between the two signals.

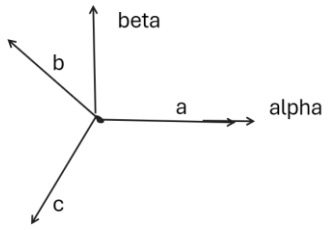
2. Clark & Park Transformation Block:

The Clark and Park transformations are mathematical transformations commonly used in the field of power electronics and control systems. These transformations are applied to convert three-phase quantities into two-phase quantities, simplifying the analysis and control of three-phase systems.

-The Clarke transformation, also known as the abc to αβ transformation, is used to convert three-phase quantities (usually represented in the abc coordinate system) into two-phase quantities (αβ coordinate system). The transformation matrix is given by,

$$\begin{bmatrix} \alpha \\ \beta \\ 0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

The αβ frame is a stationary frame but the vector $V = V_\alpha + V_\beta$ is a rotating vector. The transformation process is shown below: -



The output waveform of the control block is shown below: -

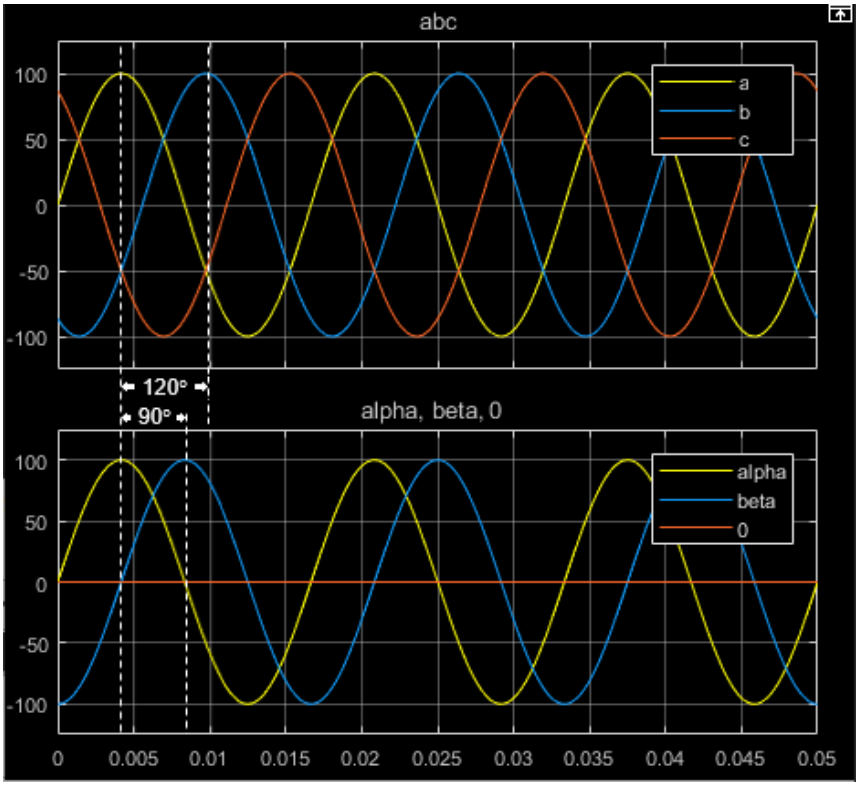


Figure 8 abc to alpha-beta

The Park transformation, also known as the αβ to dq transformation, is typically applied after the Clarke transformation. It further simplifies the two-phase representation by aligning the α axis with the rotor flux in a rotating reference frame (dq frame). The transformation matrix is given by,

$$\begin{bmatrix} i_d \\ i_q \\ i_o \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} X \begin{bmatrix} i_\alpha \\ i_\beta \\ i_o \end{bmatrix}$$

The Θ in the transformation matrix represents the angle of rotation of the dq frame relative to the αβ frame. The resulting d and q components represent the direct and quadrature components of the original αβ system.

The Park transformation is particularly useful in the control of synchronous machines, as it simplifies the representation of the machine's electrical variables in a coordinate system that rotates with the rotor flux. The transformation process is shown in the figure below.

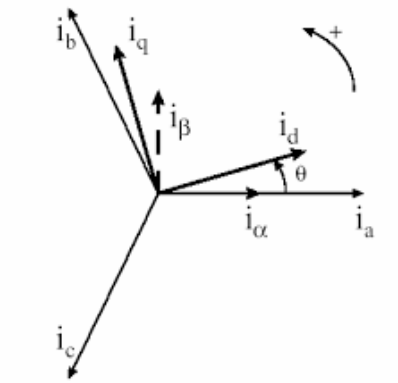
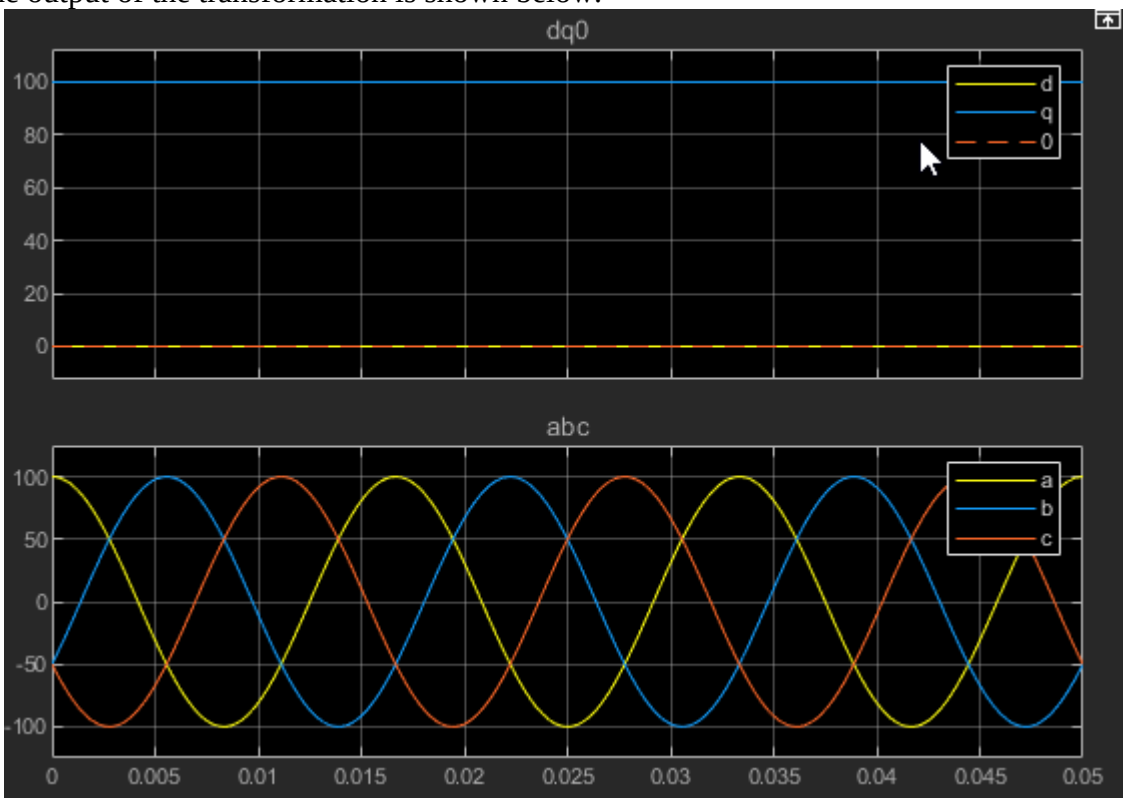


Figure 9 Park Transformation

The output of the transformation is shown below.



Buck-Boost Controller:

The function of this controller is to generate PWM signals that will decide whether buck or boost should operate at any given time. The block diagram of this controller is given below.

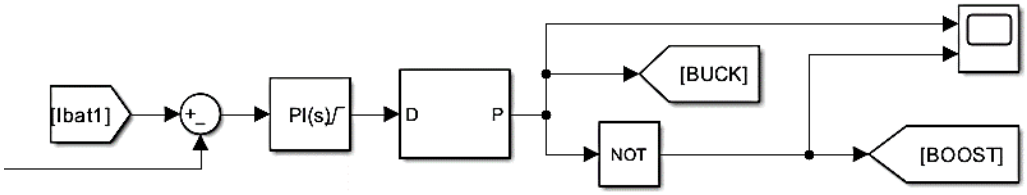


Figure 10: Buck-Boost Controller

Here the battery current is iteratively adjusted to be equal to a reference value. When the reference current is positive, the controller generates PWM signals that will cause battery to discharge and when it is negative, it will charge. The output of this block is given below.

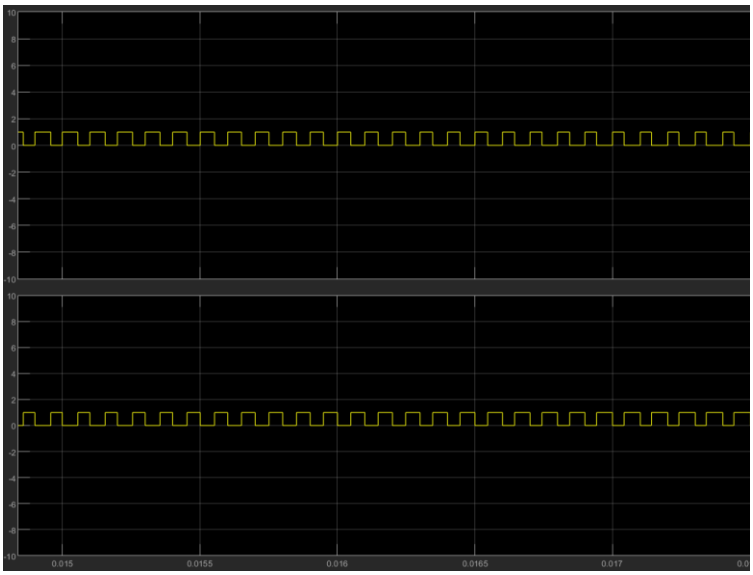


Figure 11: PWM Signal for Buck and Boost

We can see that when buck is on, boost is off and when boost is on buck is off.

Variable Load Demand:

We have modeled the varying load demand seen in households by connecting parallel loads that are connected to the grid via switches. The switches are controlled using pulse signals with period of 5sec and duty cycle of 50%. Meaning the loads will be connected to the grid for 2.5sec and will be disconnected form the grid for the rest of the 2.5sec. The block is shown below.

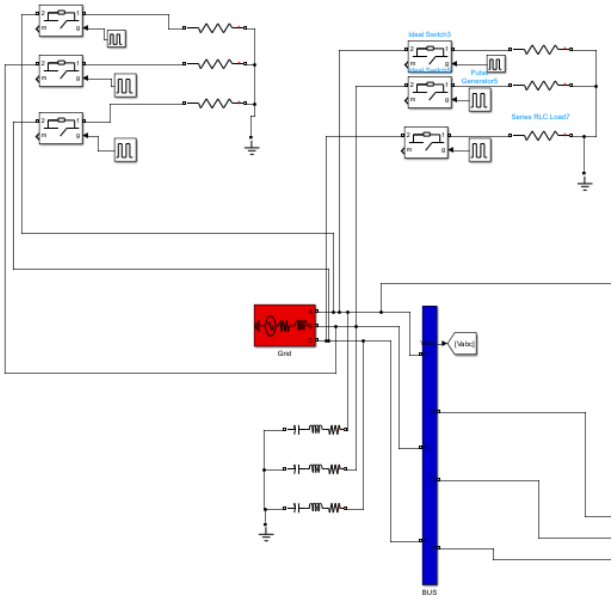


Figure 12 Varing Load

Inverter Controller:

This block generates the PWM signals that will be fed into the inverter block. A cascade control in synchronous reference frame is proposed for the inverter controller. The conventional standard vector control using 4 PI controllers in a nested loop is shown in Fig.10 . The control structure consists of two outer voltage control loops and two inner current control loops. The d-axis outer loop controls the dc bus voltage and inner loop controls the active ac current. Similarly, the q-axis outer loop regulates the ac voltage magnitude by adjusting the reactive current, which is controlled by the q-axis inner current loop. Also, dq decoupling terms L and feed-forward voltage signals are added to improve the performance during transients.

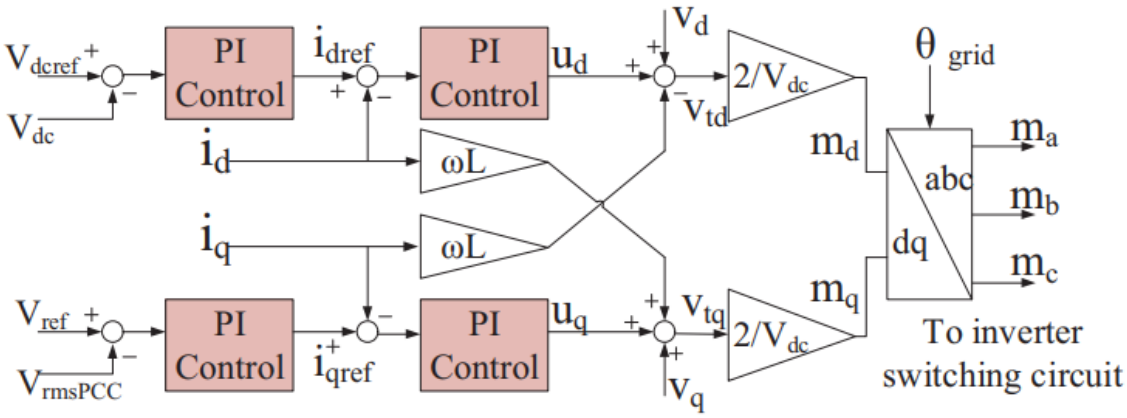


Figure 13: Inverter Controller

Our Simulink model is shown below.

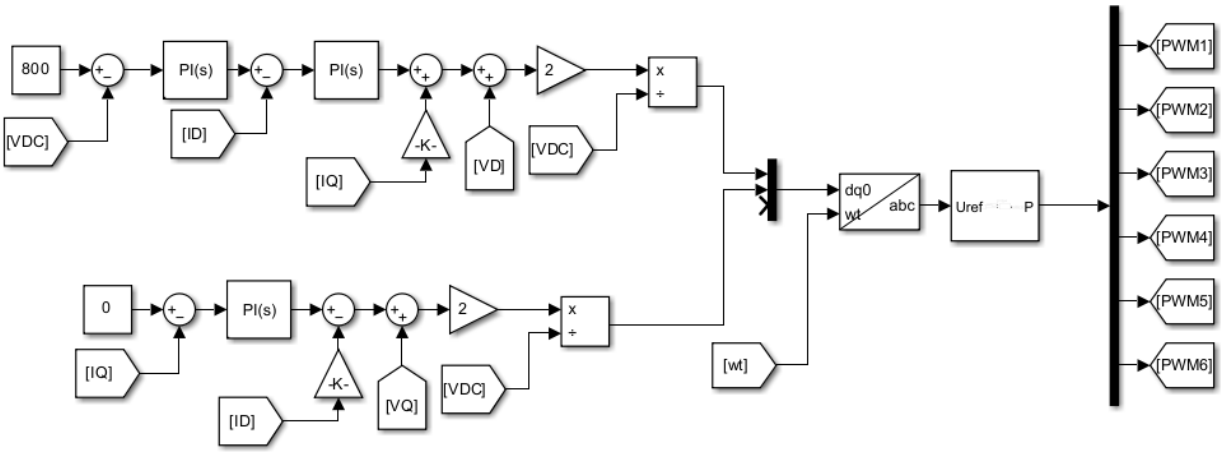


Figure 14: Simulink Model of inverter Controller.

5 Design Analysis and Evaluation

5.1 Novelty

Our project explored the behavior of a microgrid under varying load conditions, drawing inspiration from the base paper "Vehicle-To-Grid Technology in a Microgrid Using DC Fast Charging Architecture." We simulated the addition of extra loads to the grid, observing how this increased current draw impacted the system's voltage stability.

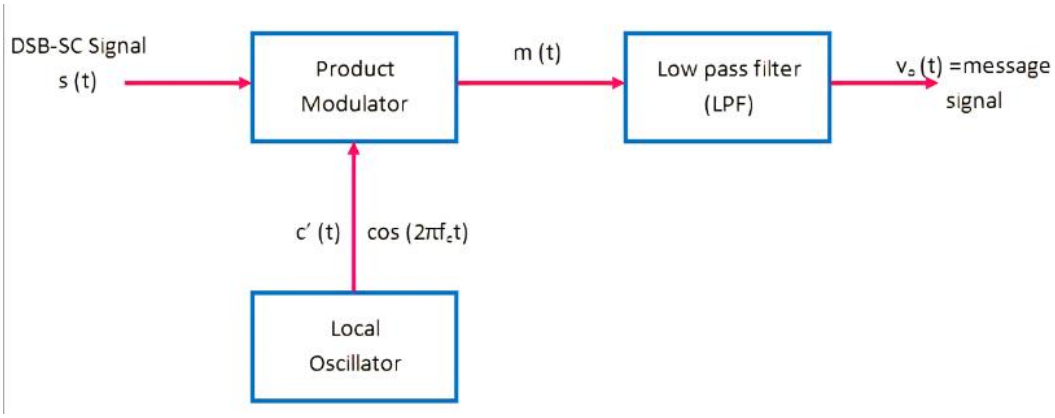
As anticipated, the increased current flow led to resistive losses within the transmission lines and transformers, mirroring how friction reduces water flow in a pipe. These losses manifested as a measurable voltage drop across the grid. To maintain constant power within the system, the voltage decrease was directly proportional to the additional load.

To manage the dynamic nature of the load, we implemented an ideal switch and pulse control system. This allowed for precise load switching, ensuring efficient utilization of available power while maintaining grid stability. Additionally, we incorporated a three-phase RLC load into the simulation, further enriching the complexity and realism of the model.

However, our primary focus was on developing a novel technique for controlling the charging and discharging of an electric vehicle (EV) based on the grid's voltage fluctuations. We deviated from conventional methods and adopted a unique approach inspired by coherent detection, a well-established technique in the communication systems domain.

The process involved:

1. **Envelope Detection:** We multiplied the single-phase voltage signal with a sin wave of the same frequency, essentially extracting the envelope of the voltage waveform. This step provided a clear indication of the voltage fluctuations without the high-frequency components.



2. **Threshold Setting:** A carefully chosen threshold voltage level was set. When the detected envelope voltage exceeded this threshold, it indicated sufficient grid capacity, and the EV initiated charging. Conversely, if the envelope voltage dipped below the threshold, signifying insufficient grid capacity, the EV switched to discharging mode, supporting the grid by injecting stored energy.

This novel control technique, based on envelope detection and thresholding, offered a dynamic and efficient way to integrate an EV into the microgrid, ensuring both grid stability and optimal utilization of the vehicle's battery.

5.2 Design Considerations (PO(c))

In our project, we have carefully considered various design aspects to optimize system performance and efficiency. One key decision was the adoption of the PV bus architecture over the swing bus configuration. Unlike the swing bus, which represents a generator bus with fixed voltage magnitude and variable phase angle, the PV bus provides control over both voltage magnitude and phase angle. This enhanced flexibility allows for better representation of real-world scenarios, facilitating improved voltage regulation and reactive power control. Moreover, PV buses support the seamless integration of renewable energy sources like solar and wind, enabling operators to dynamically adjust voltage levels to maximize energy production. This is particularly beneficial in systems with high renewable energy penetration, where PV buses enhance resilience to voltage fluctuations and grid disturbances, thereby bolstering overall grid reliability. Additionally, to maintain constant voltage levels, we opted for LCL filters over other filtering options such as RC filters. Unlike resistive elements, which would dissipate power and act as a load, LCL filters provide effective harmonic mitigation without compromising power efficiency. Furthermore, our design incorporates buck-boost converters to ensure stable output voltage across a wide input voltage range, making them ideal for battery-powered applications where maximizing battery life is paramount. The compact and versatile nature of buck-boost converters simplifies system design and reduces component count, offering space-efficient solutions for diverse applications ranging from portable devices to automotive electronics. Complementing these elements, inverters have been integrated into our project to enable seamless conversion of DC power to AC, facilitating compatibility with various electrical systems and appliances. Overall, our design approach prioritizes efficiency, flexibility, and reliability to meet the evolving demands of modern power systems.

When the load was added, we observed a transient in the system due to the switching action associated with the load connection process. Transients occur during load switching primarily because of the sudden change in the system's electrical characteristics, such as voltage and current levels. When a load is connected or disconnected, there is an instantaneous alteration in the load impedance, resulting in a disturbance in the system's equilibrium. This abrupt change induces transient effects, including voltage spikes or dips, current surges, and oscillations, as the system responds to the new operating conditions. Additionally, the switching process itself introduces disturbances, such as electromagnetic interference and voltage transients, further exacerbating the transient response. Therefore, the observed transient during load addition can be attributed to the combined effects of load impedance variations and switching-induced disturbances, highlighting the importance of carefully managing load connections to minimize transient effects and ensure system stability.

We opted for coherent detection over envelope detection due to its superior performance and robustness in signal-processing applications. Coherent detection involves directly recovering the phase and frequency of the modulated signal, enabling precise demodulation even in the presence of noise and interference. This method preserves the phase relationship between the carrier and the modulating signal, allowing for accurate detection and reconstruction of the original signal. In contrast, envelope detection relies solely on extracting the amplitude variations of the modulated signal, making it more susceptible to noise and distortion, especially in low-SNR (Signal-to-Noise Ratio) conditions. By leveraging coherent detection, we ensure high fidelity and reliability in signal recovery, essential for demanding applications

such as communication systems, radar, and instrumentation, where accurate signal demodulation is critical for optimal performance.

5.2.1 Results and Analysis

The battery voltage, current and State of charge is given below:

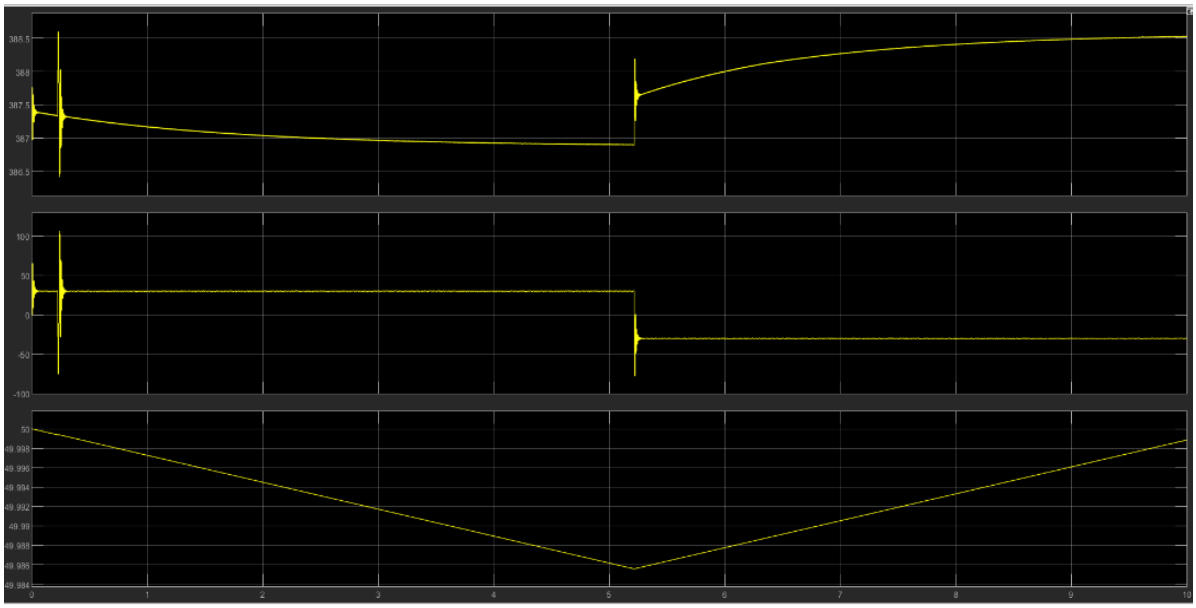


Figure 15 Battery Condition

The grid voltage is given below:

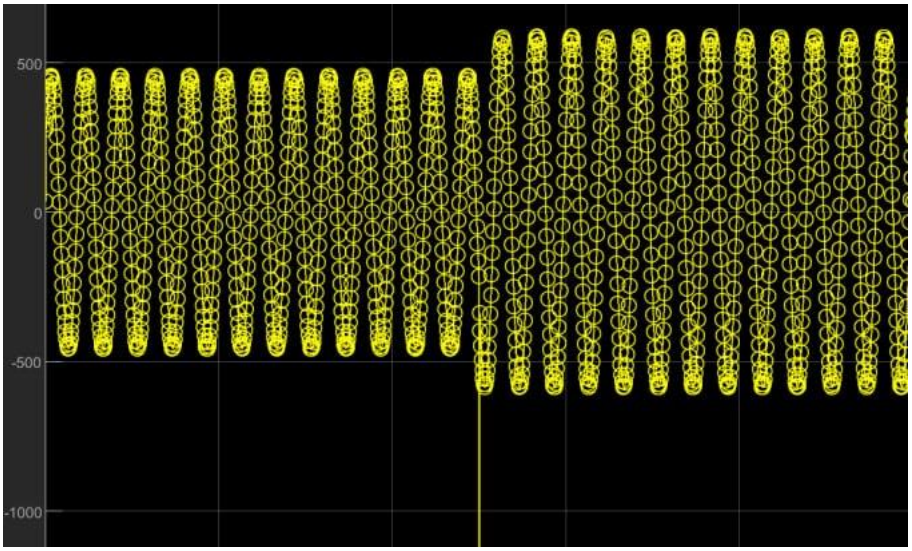


Figure 16 Grid Voltage

If we look closely at the 2nd graph then initially load is higher in the grid . so the grid voltage is low. At this time it is expected to get current from the battery. If we look at the battery condition then we can see that voltage of the battery is decreasing and the state of charge also indicates that which means V2G mood is operating. Again when load is less in the grid then the grid voltage becomes high and at that point G2V mood is working. That's why the voltage of the battery is increasing and

state of charge is also increasing.

5.3 Limitations of Tools (PO(e))

Limitations of implementing V2G and G2V projects solely in MATLAB Simulink without real-world deployment include:

1. Lack of validation in real-world conditions may lead to discrepancies between simulated results and actual performance.
2. Inability to account for physical constraints and environmental factors present in real-world settings could affect the accuracy of simulations.
3. Limited understanding of higher-level theoretical concepts related to V2G and G2V systems may constrain the depth of analysis within Simulink models.
4. Absence of practical experience in deploying V2G and G2V technologies may hinder the identification and resolution of implementation challenges.
5. Dependency on Simulink's simulation capabilities may overlook nuances and complexities inherent in real-world interactions between vehicles and the grid.
6. Insufficient integration with external data sources and models may restrict the ability to incorporate real-time information and dynamic conditions.
7. Lack of hands-on experience with hardware components and systems may impede the translation of simulation results into actionable insights.
8. Limited opportunities for experimentation and iteration in a controlled environment may inhibit learning and innovation.
9. Challenges in translating theoretical knowledge into practical solutions may hinder the development of robust V2G and G2V implementations.
10. Without real-world deployment, the full potential and applicability of V2G and G2V solutions may remain unexplored, limiting their impact on sustainability and energy management efforts.

5.4 Sustainability Evaluation (PO(g))

In case of environmental impact assessment, we can quantify the reduction in greenhouse gas emissions achieved through V2G and G2V systems compared to conventional vehicle usage and grid operations. Then we can analyze the potential for integrating renewable energy sources and reducing reliance on fossil fuels.
Social Impact Analysis: We have to evaluate the accessibility and affordability of V2G and G2V technologies across different socioeconomic groups. It assess the potential for job creation in renewable energy and electric vehicle sectors. We have to nalyze the impact of V2G and G2V systems on energy poverty and access in underserved communities.
Then we can Conduct a cost-benefit analysis comparing upfront investment and operational costs with long-term economic benefits.
Technological Viability: We can evaluate reliability and performance under various conditions, including grid demand fluctuations and vehicle usage patterns.

This structured framework provides a comprehensive approach to evaluating the sustainability of V2G and G2V projects, ensuring that environmental, social, economic, and technological aspects are systematically assessed and addressed.

6 Reflection on Individual and Team work (PO(i))

6.1 Logbook of Project Implementation & Individual Contribution

Milestone achieved	Individual Role	Team Role
Grid design	2006145	Everyone helped one another to do their individual task
LCL Filter	2006133	
Inverter Design	2006153	
Inverter Controller Design	2006134	
Buck Boost Controller & Battery	2006136	

7 Future Work (PO(l))

While this project has made significant strides in exploring the potential of Vehicle-to-Grid (V2G) and Grid-to-Vehicle (G2V) technologies within micro-grid environments, several avenues for future research and development warrant consideration. The following areas represent promising directions for further investigation and innovation:

1. **Optimization of Charging Infrastructure**: Future work could focus on refining the design and implementation of charging infrastructure to enhance efficiency, reliability, and scalability. This may involve exploring advanced charging technologies, such as wireless charging systems, and optimizing charging station placement within the micro-grid network to maximize accessibility and utilization.

2. **Enhanced Control Strategies**: Developing advanced control algorithms and strategies for V2G-G2V systems is essential to optimize energy transfer, grid stability, and user experience. Future research could explore machine learning algorithms, predictive analytics, and adaptive control techniques to dynamically adjust charging and discharging schedules based on grid conditions, user preferences, and energy market dynamics.

3. **Integration of Renewable Energy Sources**: As renewable energy penetration continues to grow, integrating V2G-G2V systems with renewable energy sources presents an exciting opportunity to enhance sustainability and resilience. Future work could focus on optimizing the integration of solar, wind, and other renewable resources with EV charging infrastructure, leveraging smart grid technologies to balance energy supply and demand in real-time.

4. **Standardization and Interoperability**: Establishing industry-wide standards and protocols for V2G-G2V systems is crucial to ensure interoperability, compatibility, and seamless integration with existing infrastructure. Future research could contribute to the development of standardized communication protocols, interoperable hardware

interfaces, and regulatory frameworks to facilitate widespread adoption and deployment of V2G-G2V technologies.

5. **Lifecycle Assessment and Environmental Impact**: Conducting comprehensive lifecycle assessments and environmental impact studies of V2G-G2V systems is essential to evaluate their sustainability and quantify their environmental benefits. Future research could investigate the lifecycle environmental impacts of EV batteries, charging infrastructure, and grid interactions, considering factors such as manufacturing, transportation, operation, and end-of-life disposal.

6. **Real-world Deployment and Field Trials**: Transitioning from simulation studies to real-world deployment and field trials is critical to validate the feasibility, scalability, and performance of V2G-G2V systems in practical settings. Future work could involve collaborating with industry partners, utilities, and transportation agencies to pilot V2G-G2V projects in diverse geographical locations and demographic contexts, gathering empirical data and user feedback to inform system optimization and refinement.

In conclusion, the future of V2G-G2V technologies within micro-grid environments holds immense promise for transforming the energy landscape, enhancing sustainability, and empowering communities to actively participate in energy management. Continued research, innovation, and collaboration across academia, industry, and government sectors will be essential to unlock the full potential of V2G-G2V systems and realize a more resilient, decentralized, and sustainable energy future.

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